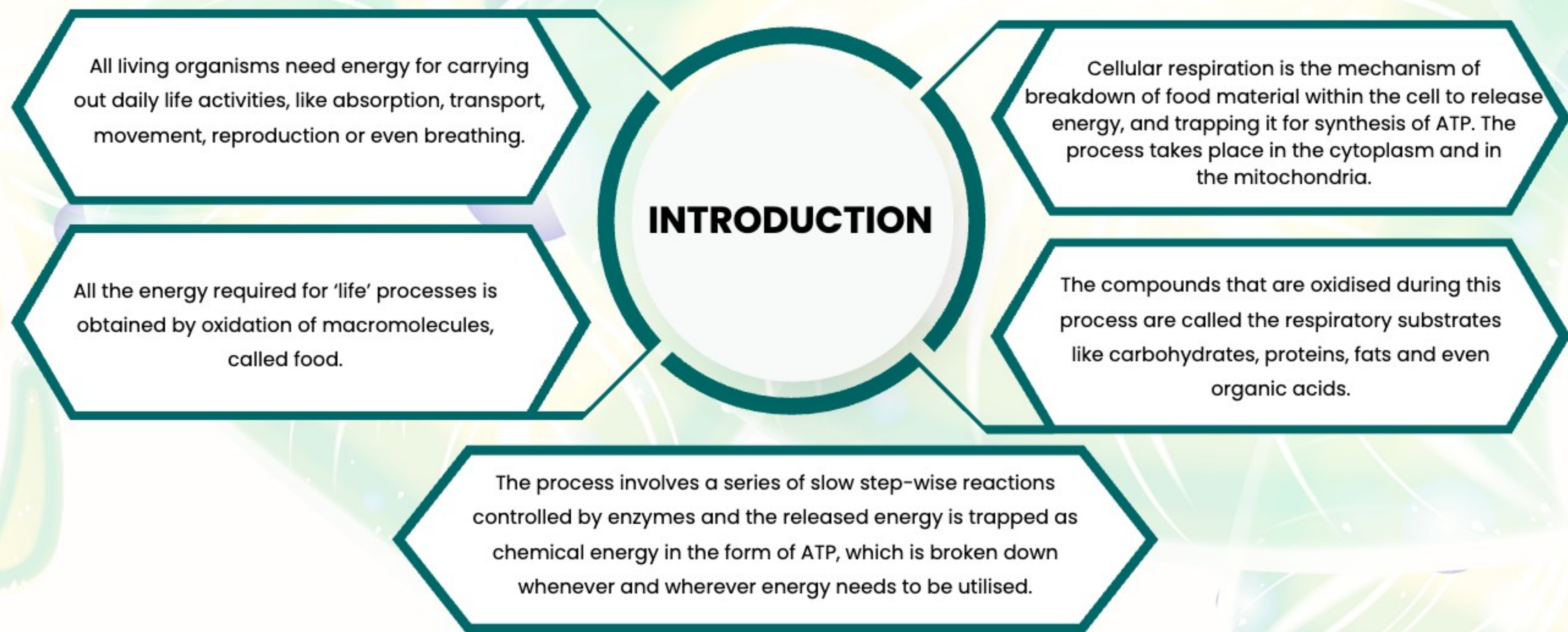




RESPIRATION IN PLANTS



DO PLANTS BREATHE?

Plants have systems in place to ensure O_2 availability, i.e. stomata and lenticels for this purpose.

Each plant part takes care of its own gas exchange needs. There is very little transport of gases from one plant part to another.

Roots, stems and leaves respire at rates far lower than animals do.

Most cells of a plant have at least a part of their surface in contact with air.

Complete combustion of glucose produces CO_2 and H_2O as end products and yields energy most of which is given as heat
 $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$

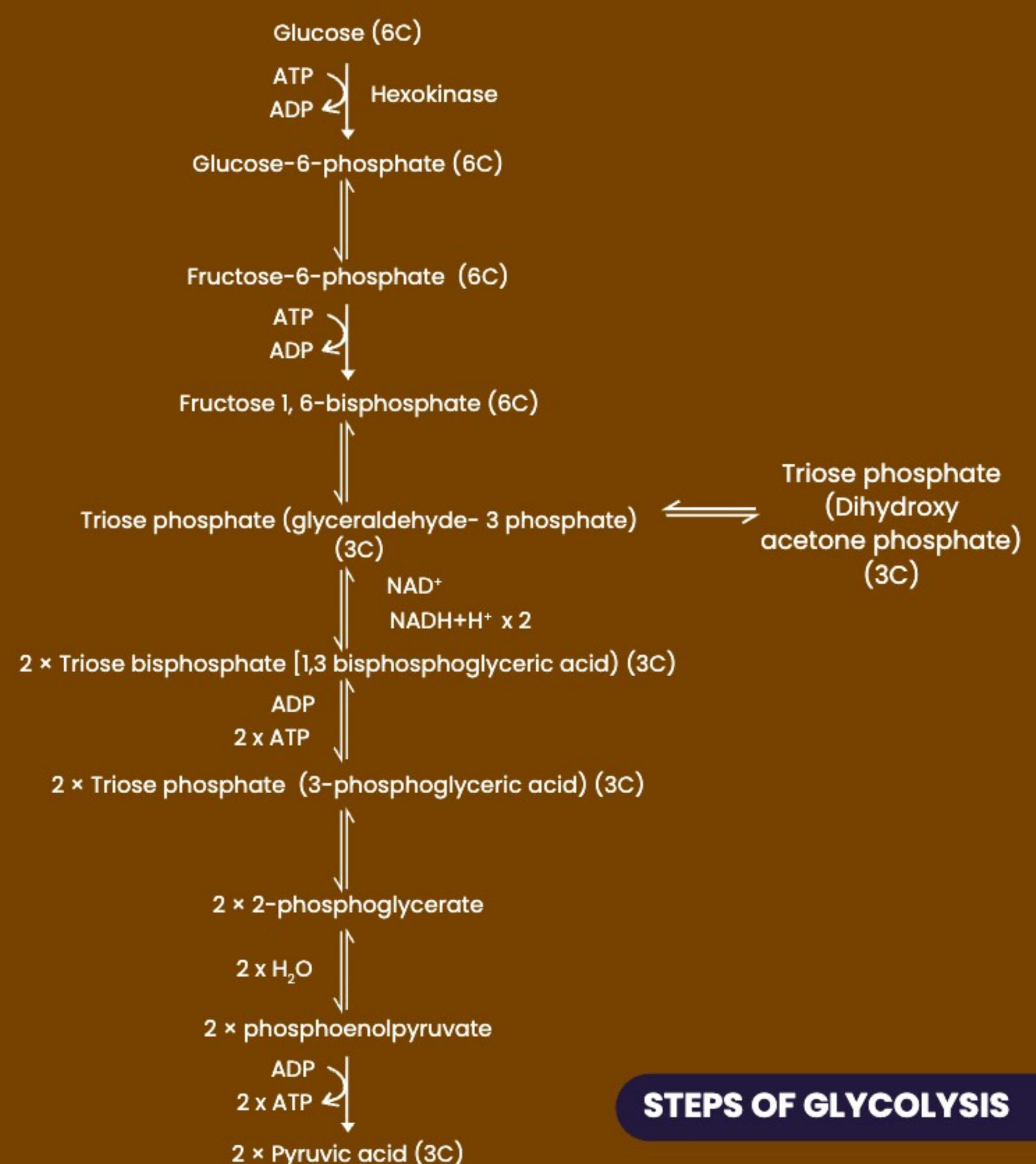
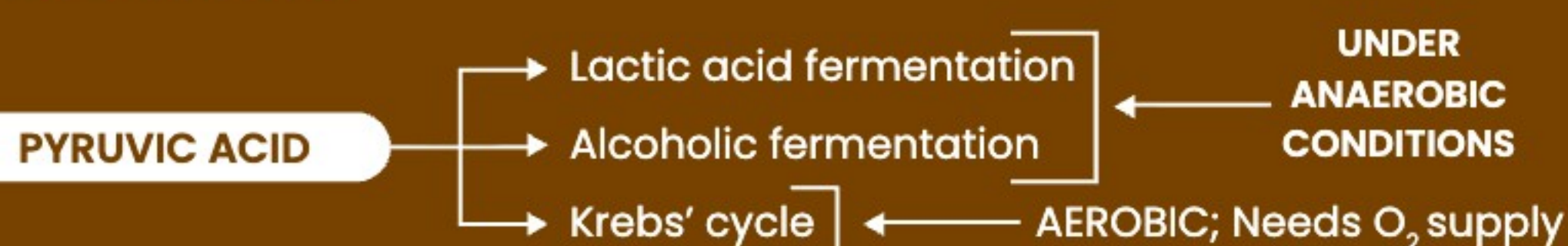
But plants oxidise glucose in several small steps and energy released is coupled to ATP synthesis.

Facultative and obligate anaerobes, can respire in absence of O_2 .

All organisms retain this strategy of partial glucose oxidation in absence of oxygen called GLYCOLYSIS.

GLYCOLYSIS

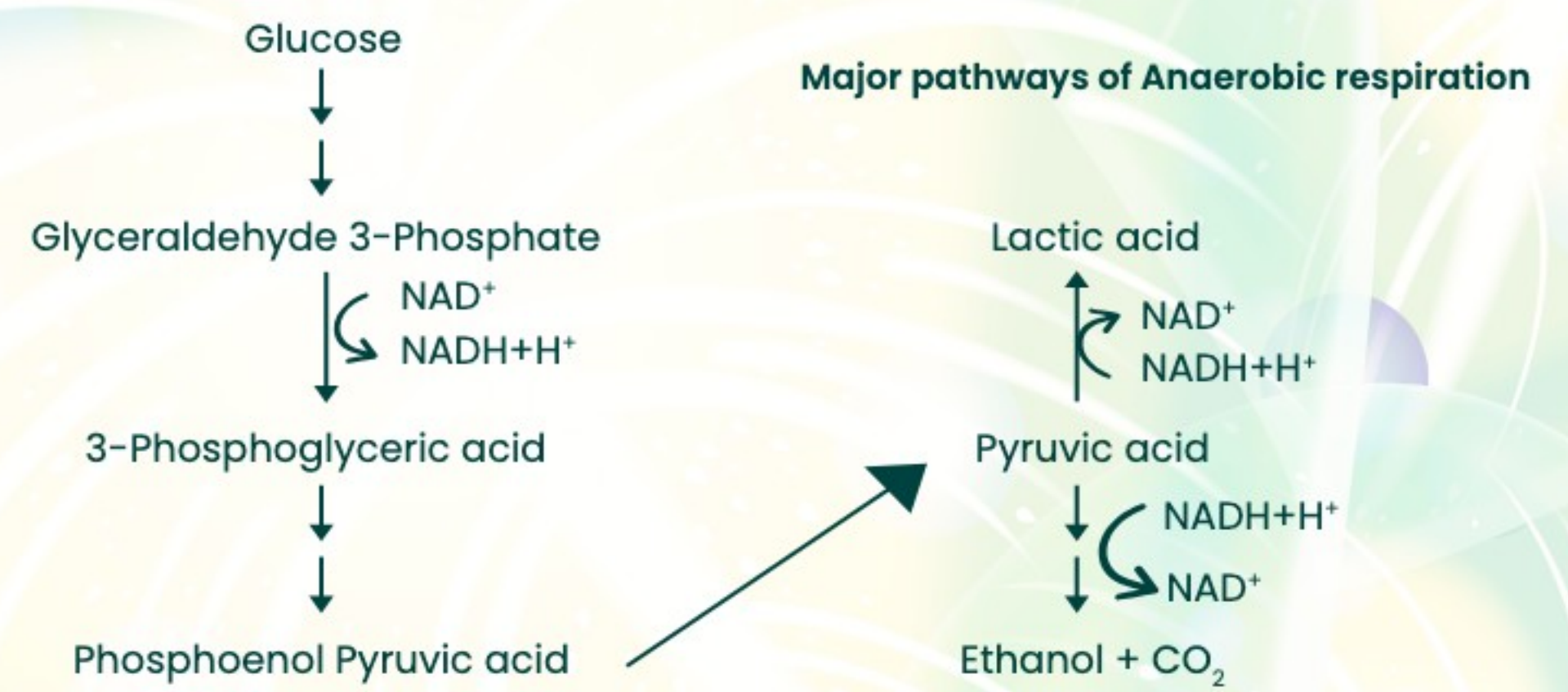
- Greek-*glycos* - sugar and *lysis* = splitting
- Scheme given by Embden, Meyerhof and Parnas. referred as EMP- pathway
- In anaerobic organisms, it is the only process in respiration
- Occurs in cytoplasm and present in all living organisms.
- In this process glucose undergoes partial oxidation to form two molecules of pyruvic acid.
- In plants, glucose comes from sucrose (the end product of photosynthesis) or from storage carbohydrates
- Sucrose is converted into glucose and fructose by invertase and these monosaccharides readily enter the glycolytic pathway
- In glycolysis, a chain of ten reactions produces pyruvate from glucose by the help of different enzymes.
- In glycolysis 2 ATP are utilised and total 4 ATP, 2 $NADH+H^+$ and 2 molecules of pyruvic acid are produced.
- Pyruvic acid is the key product of glycolysis and its metabolic fate depends on cellular need.





FERMENTATION

- In fermentation, by yeast, incomplete oxidation of glucose is achieved under anaerobic conditions to produce CO_2 and ethanol, by the help of enzymes pyruvate decarboxylase and alcohol dehydrogenase.
- Bacteria and in animal cells (muscles during exercise, when oxygen is inadequate for cellular respiration) pyruvate is reduced to lactic acid by lactate dehydrogenase.
- Less than 7% of the energy in glucose is released
- Also the processes are hazardous as either acid or alcohol is produced.
- Yeast poison themselves to death when concentration of alcohol reaches about 13%.



AEROBIC RESPIRATION

1. In eukaryotes, it takes place in mitochondria. Leads to complete oxidation of organic substances, in the presence of oxygen and releases CO_2 , water and a large amount of energy present in the substrate.

2.

This type of respiration is most common in higher organisms

3. For aerobic respiration to take place within mitochondria, the final product of glycolysis is transported into mitochondria from cytoplasm

4.

Crucial events of aerobic respiration are
 a. Complete oxidation of pyruvic acid $\rightarrow$ site = Mitochondrial matrix
 b. ETS and synthesis of ATP $\rightarrow$ site = Inner mitochondrial membrane.

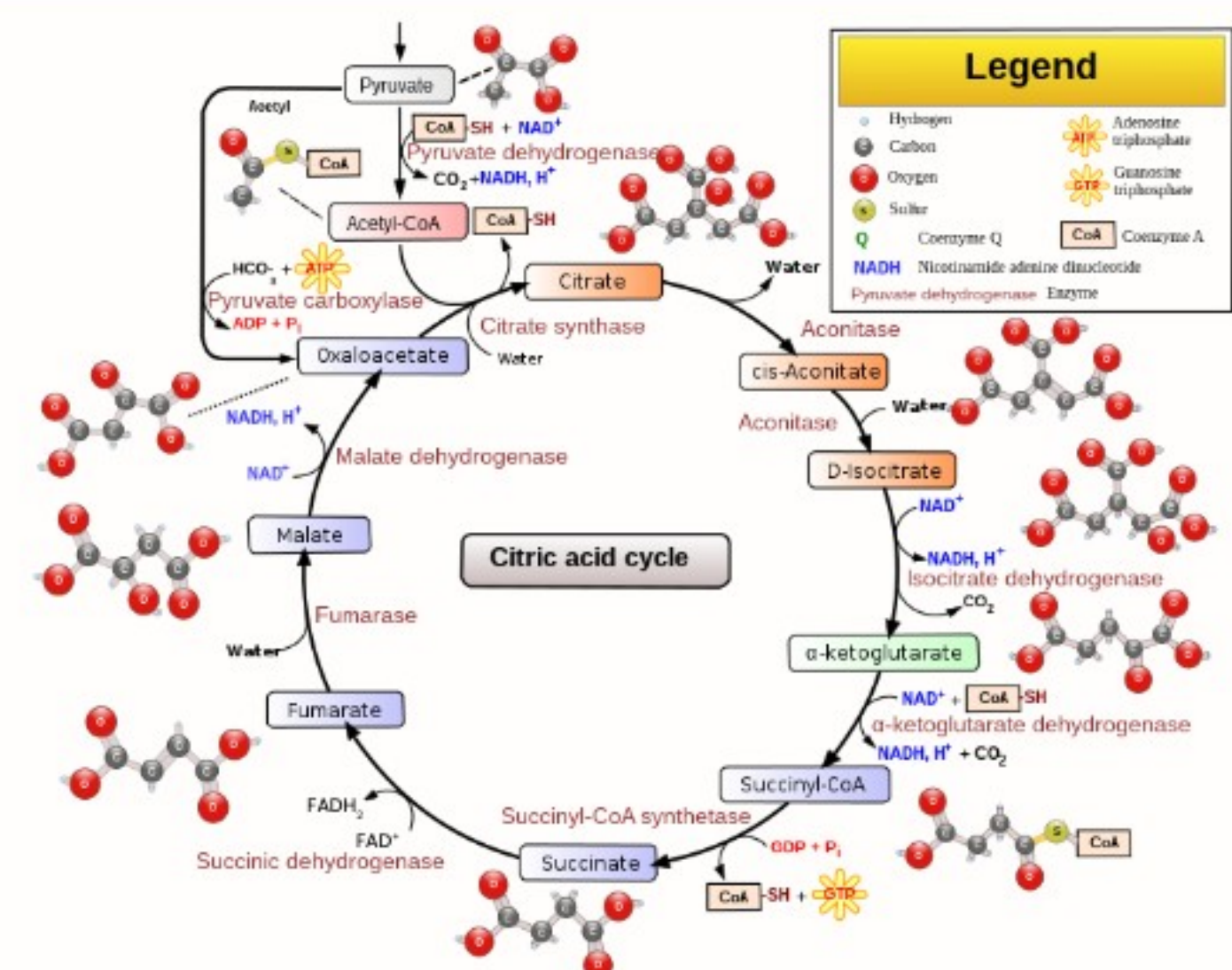
TRICARBOXYLIC ACID CYCLE (TCA cycle or Krebs' cycle) (In mitochondrial matrix)

- Acetyl CoA produced by oxidative decarboxylation of pyruvic acid enters the TCA cycle more commonly known as Krebs' cycle. (Scientist $\rightarrow$ Hans Krebs)
- First reaction of Krebs' cycle is condensation, then isomerisation.

$$\text{OAA} \xrightarrow[\text{Citrate synthase}]{\text{AcetylCoA}} \text{Citric Acid} \xrightarrow{\text{Isomerises}} \text{Isocitrate}$$
- Followed by two successive decarboxylation to form α -ketoglutarate and the succinyl CoA
- During conversion of succinyl CoA to succinic acid, substrate level phosphorylation takes place to produce GTP which in a coupled reaction simultaneously produces ATP.
- The summary equation for this phase of respiration is:

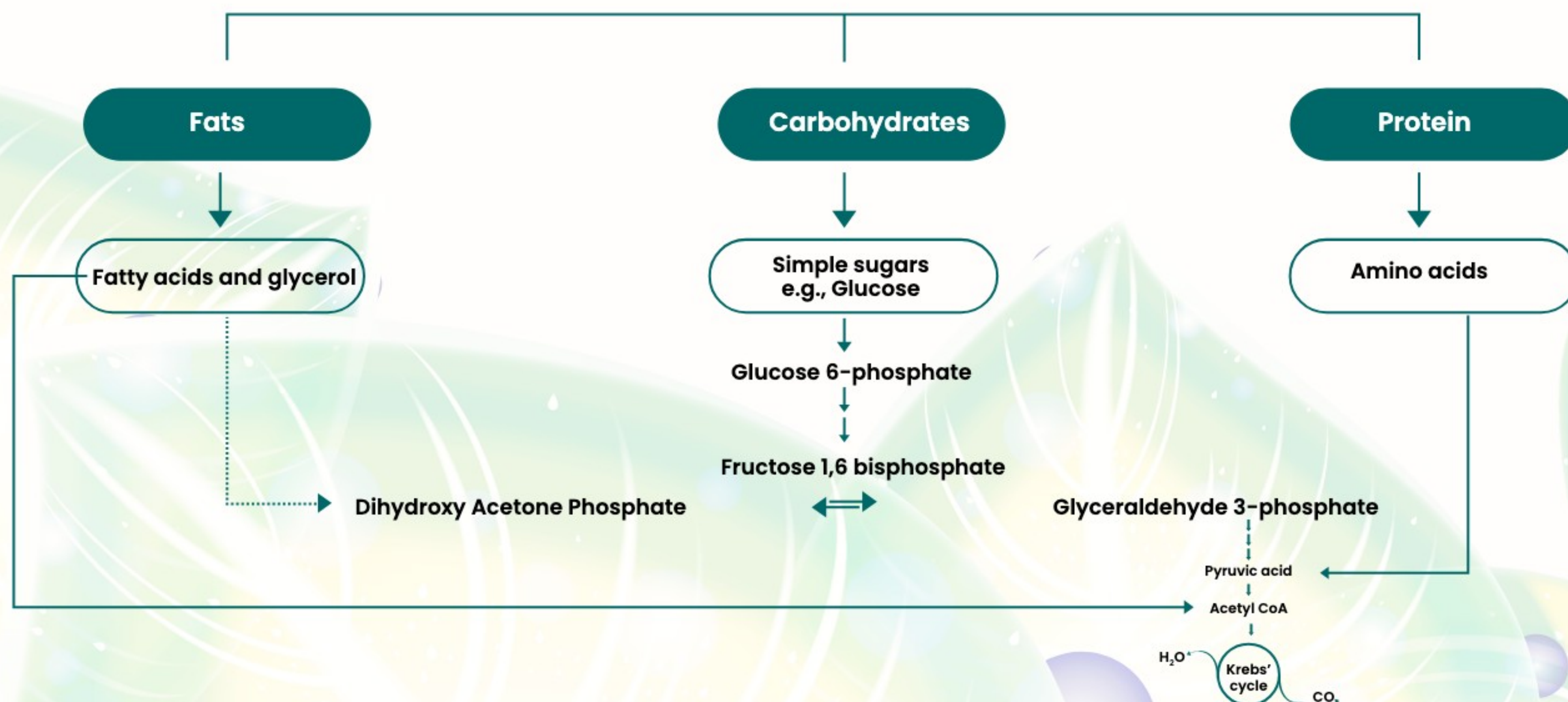
$$\text{Pyruvate} + 4\text{NAD}^+ + \text{FAD} + 2\text{H}_2\text{O} + \text{ADP} + \text{P}_i$$

$$\xrightarrow[\text{MATRIX}]{\text{MITOCHONDRIAL}} 3\text{CO}_2 + 4\text{NADH} + 4\text{H}^+ + \text{FADH}_2 + \text{ATP}$$
- So, per molecule of glucose, 8 $\text{NADH} + \text{H}^+$, two FADH_2 , and 2 ATP are synthesised from pyruvic acid.



AMPHIBOLIC PATHWAY

Respiratory pathway is involved in both anabolism and catabolism hence it is called amphibolic pathway.

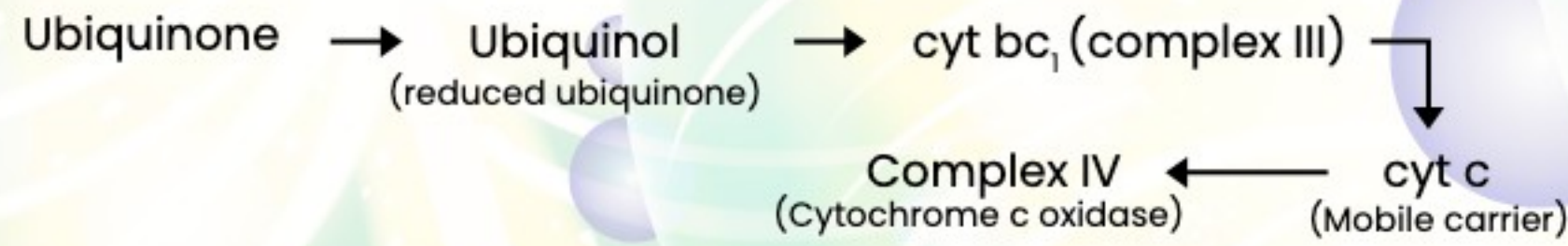




ELECTRON TRANSPORT SYSTEM (ETS) AND OXIDATIVE PHOSPHORYLATION

• $\text{NADH} + \text{H}^+$ and FADH_2 are oxidised through ETS and the electrons are passed on to O_2 resulting in formation of H_2O through various complexes in the inner mitochondrial membrane.

• NADH dehydrogenase (Complex-I) and FADH_2 (Complex-II) transfers electrons to



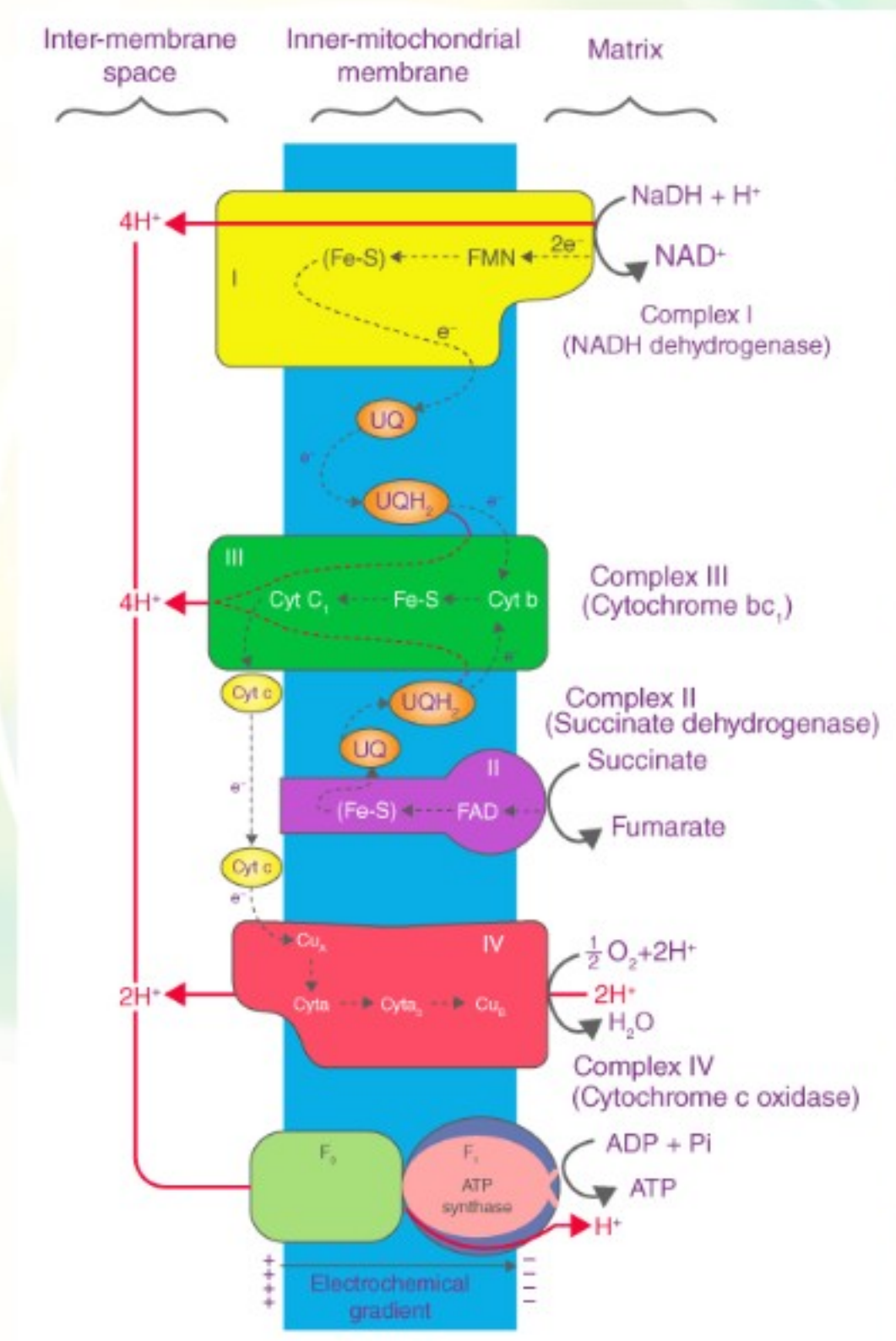
• When electrons pass from one carrier to another via complex-I to IV in ETC. they are coupled to ATP synthase (complex-V) for production of ATP from ADP and inorganic phosphate

• Oxidation of one NADH gives 3 ATP, while one FADH_2 gives two ATP.

• The role of oxygen is limited to the terminal stage. Yet the presence of oxygen is vital, since it drives the whole process by removing hydrogen from the system. Oxygen is the final acceptor of hydrogen and electrons.

• Complex-V has two major components, $\text{F}_0\text{-F}_1$ is peripheral membrane protein complex and contains site for ATP synthesis and F_0 forms the channel through which protons cross the inner membrane.

The passage of protons through the channel is coupled to the catalytic site of F_1 for production of ATP. For each ATP produced. 2H^+ passes through F_0 from the intermembrane space to matrix down the electrochemical proton gradient



RESPIRATORY BALANCE SHEET

(In reality it is a theoretical exercise as all pathways work simultaneously and do not take place one after another. Enzymatic rates are controlled by multiple means.)

There can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of glucose

In fermentation there is net gain of only 2 ATP for each molecule of glucose degraded

NADH is oxidised to NAD^+ slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration.

RESPIRATORY QUOTIENT = (RQ)

01

The ratio of volume of CO_2 evolved to the volume of O_2 consumed is RQ.

$$\text{RQ} = \frac{\text{Volume of } \text{CO}_2 \text{ evolved}}{\text{Volume of } \text{O}_2 \text{ consumed}}$$

02

It depends on the type of respiratory substrate, used during respiration

03

For, carbohydrates = 1
Fat = less than 1. (eg-tripalmitin=0.7)
Protein = about 0.9

• Breaking of C-C bonds of complex organic molecules leads to release of lot of energy in cellular respiration.

• Glucose is the preferred substrate, though fats and protein can also yield energy

• Fermentation takes place in many prokaryotes, unicellular eukaryotes and in germinating seeds

• In aerobic respiration O_2 is ultimate electron acceptor and it gets reduced to water